

## 4 First-Order Scheme

### 4.3 Numerical experiments

We assess the first-order discretization by a manufactured-solution test on  $\Omega = (0, 1)^3$  with  $T = 2$ . Each Cartesian partition into  $K \times K \times K$  cubes is subdivided into tetrahedra, with  $K = 4, 8, 16, 32$  and  $\Delta t = 1/K = h/\sqrt{3}$ , where  $h$  is the maximum tetrahedron diameter. The same test is used for the second-order scheme below. Space and time are refined together; this experiment does not isolate the temporal convergence order.

**Manufactured solution.** Define

$$q(s) = 64s^3(1-s)^3, \quad B(x, y, z) = q(x)q(y)q(z), \quad S(x, y, z) = \sin^2(\pi x) \sin^2(\pi y) \sin^2(\pi z),$$

and let  $a = (1, 2, 3)^\top$ ,  $b = (1, -1, 1/2)^\top$ , and  $c = (1, 2, -1)^\top$ . The prescribed fields are

$$\begin{aligned} \mathbf{u} &= \frac{1 + 0.1t}{C_u} \nabla B \times a, & \boldsymbol{\omega} &= \frac{1 - 0.05t}{C_\omega} Bb, & \mathbf{m} &= \frac{1 + 0.05t}{C_m} Sc, \\ \varphi &= \frac{1 - 0.05t}{C_\varphi} B, & \mathbf{H} &= \nabla \varphi, & p &= \frac{1 + 0.05t}{C_p} \prod_{s \in \{x, y, z\}} \cos(2\pi s), \end{aligned}$$

where the fixed normalization constants used in the computations are

$$\begin{aligned} C_u &= 363.275188821, & C_\omega &= 20.6500992827, & C_m &= \frac{\sqrt{81 + 108\pi^2}}{16}, \\ C_\varphi &= 13.7667328551, & C_p &= 0.353553390593. \end{aligned}$$

The auxiliary fields are  $\mathbf{k} = \text{curl } \mathbf{m}$ ,  $\mathbf{z} = \mathbf{u} \times \mathbf{m}$ ,  $\vartheta = -\Delta \varphi$ , and  $\varpi = \mathbf{u} \cdot \mathbf{m}$ . We set  $\mathbf{H}_e = -\mu_0(\mathbf{m} + \mathbf{H})$  and use the zero-mean modified pressure  $\tilde{p} = \mathcal{P}_0[p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2]$ , where  $\mathcal{P}_0 v = v - |\Omega|^{-1} \int_\Omega v$ . All physical parameters  $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$  are set to one.

This is a forced verification problem. The momentum, angular-momentum, and magnetization sources  $\mathbf{f}, \mathbf{g}, \mathbf{r}$  are obtained by substituting the prescribed fields into the corresponding continuous equations. Their weak right-hand sides are  $(\mathbf{f}, \mathbf{v}_h)$ ,  $(\mathbf{g}, \mathbf{s}_h)$ , and  $(\mathbf{r}, \mathbf{F}_h)$ ; the magnetic-potential right-hand side is

$$\left( -\partial_t \mathbf{H}_e - \frac{\mu_0 + \chi_0}{\tau \mu_0} \mathbf{H}_e + \sigma \nabla \text{div } \mathbf{H}_e - \mu_0 \mathbf{r}, \nabla \psi_h \right).$$

The fields satisfy the homogeneous boundary traces used in the experiment: zero velocity and angular velocity, zero normal magnetization, zero tangential  $\mathbf{z}, \mathbf{k}, \mathbf{H}$ , and zero  $\varphi, \vartheta, \varpi$ . In particular,  $q, q', q''$  vanish at both endpoints and  $S, \nabla S$  vanish on the cube boundary. The new  $\vartheta$  and  $\varpi$  couplings are not identically zero for this test.

**Discretization and implementation.** The first-order runs use the MINI velocity element, continuous piecewise linear pressure and angular velocity, lowest-order Raviart–Thomas magnetization and first-family Nedelec  $\mathbf{z}, \mathbf{k}, \mathbf{H}$ , and continuous piecewise linear  $\varphi, \vartheta, \varpi$ . The initial velocity is obtained by a discrete Stokes projection. After interpolating the initial magnetization into its RT space, the initial potential is determined from the compatible magnetic Poisson problem. The initial  $\mathbf{k}, \mathbf{z}, \vartheta, \varpi$  satisfy their discrete mass-projection relations. Pressure is normalized by its finite-element integral mean.

Computations use FEniCSx/DOLFINx 0.10.0 [1] with PETSc and MUMPS. The five-block Algorithm 3 is evaluated through matrix-free Schur-complement actions. The  $K_1, K_3, K_5$  subproblems use MUMPS LU; the  $S_2, S_4$  systems use FGMRES with LU preconditioners. The MUMPS ordering control is ICNTL(7) = 7. The Schur tolerances are  $10^{-10}$  relative and  $10^{-12}$  absolute. All first-order cases and the second-order cases up to  $K = 16$  use 48 MPI processes with one thread each; the second-order  $K = 32$  case uses 32 MPI processes with four threads each on a 1-TB-class node. Source terms are evaluated at quadrature points, with assembly quadrature degree 12. In the error calculations, the analytic values and derivatives are independently represented in a degree-6 DG space and integrated with degree-12 quadrature; the numerical solution is evaluated in its original finite-element space.

As implementation checks, the two schemes were compared with a monolithic solve of the same matrix at the first two time levels on  $K = 4$ , modulo the same pressure nullspace; the normalized solution differences were below  $1.1 \times 10^{-12}$ . An independent transcription of the nine weak equations on  $K = 2$ , using random finite-element fields and both unit and non-unit parameters, also agreed with the production matrices and right-hand sides to scaled differences below  $1.1 \times 10^{-16}$ . These checks concern algebraic and implementation consistency, rather than replacing the convergence test.

**Error measures.** For a field  $v$  and its specified norm  $X$ , we report

$$E_X(v) = \frac{\|v - v_h\|_X}{\|v\|_X}, \quad \text{rate}(K \rightarrow 2K) = \log_2 \frac{E_X(v; K)}{E_X(v; 2K)}.$$

The norms are full  $H^1$  for  $\mathbf{u}, \boldsymbol{\omega}, \varphi$ ,  $L^2$  for  $\tilde{p}, \mathbf{k}$ ,  $H(\text{div})$  for  $\mathbf{m}$ , and  $H(\text{curl})$  for  $\mathbf{H}$ . Thus the graph norms combine the squared  $L^2$  and derivative terms before taking the square root. Pressure errors use zero-mean numerical and exact fields. First-order errors are evaluated at  $T$ . We also report  $C_H$ , the maximum of  $|\text{curl } \mathbf{H}_h|$  over the degree-6 DG interpolation points, as a sampled estimate of  $\|\text{curl } \mathbf{H}_h\|_{L^\infty}$ . It is an absolute diagnostic, not a relative error or a rigorous continuous supremum.

**First-order results.** Table 1 gives the errors and observed orders; Figure 1 shows the corresponding log–log error curves. On the finest refinement interval, the rates for  $\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \mathbf{k}, \varphi, \mathbf{H}$  are 1.019, 0.994, 0.999, 0.997, 1.018, and 1.017, respectively. The modified pressure exhibits a rate of 2.065 in this test. The sampled curl diagnostic remains between  $1.22 \times 10^{-13}$  and  $8.51 \times 10^{-12}$ , consistent with the discrete gradient recovery of the magnetic field up to numerical error.

Table 1: Relative errors and observed rates for the first-order scheme,  $T = 2$  and  $\Delta t = 1/K$ .  $C_H$  is the sampled absolute curl diagnostic.

| (a) Velocity, pressure, angular velocity, and magnetization |                         |       |                         |       |                            |       |                             |       |
|-------------------------------------------------------------|-------------------------|-------|-------------------------|-------|----------------------------|-------|-----------------------------|-------|
| $K$                                                         | $\mathbf{u}: H^1$       |       | $\tilde{p}: L^2$        |       | $\boldsymbol{\omega}: H^1$ |       | $\mathbf{m}: H(\text{div})$ |       |
|                                                             | Error                   | Rate  | Error                   | Rate  | Error                      | Rate  | Error                       | Rate  |
| 4                                                           | $9.6554 \times 10^{-1}$ | –     | $5.6019 \times 10^{-1}$ | –     | $6.4182 \times 10^{-1}$    | –     | $3.9160 \times 10^{-1}$     | –     |
| 8                                                           | $4.3070 \times 10^{-1}$ | 1.165 | $1.0260 \times 10^{-1}$ | 2.449 | $3.5671 \times 10^{-1}$    | 0.847 | $1.9873 \times 10^{-1}$     | 0.979 |
| 16                                                          | $1.9858 \times 10^{-1}$ | 1.117 | $2.1172 \times 10^{-2}$ | 2.277 | $1.8331 \times 10^{-1}$    | 0.960 | $9.9738 \times 10^{-2}$     | 0.995 |
| 32                                                          | $9.8002 \times 10^{-2}$ | 1.019 | $5.0596 \times 10^{-3}$ | 2.065 | $9.2018 \times 10^{-2}$    | 0.994 | $4.9919 \times 10^{-2}$     | 0.999 |

| (b) Auxiliary curl field, potential, magnetic field, and curl diagnostic |                         |       |                         |       |                              |       |                          |
|--------------------------------------------------------------------------|-------------------------|-------|-------------------------|-------|------------------------------|-------|--------------------------|
| $K$                                                                      | $\mathbf{k}: L^2$       |       | $\varphi: H^1$          |       | $\mathbf{H}: H(\text{curl})$ |       | $C_H$                    |
|                                                                          | Error                   | Rate  | Error                   | Rate  | Error                        | Rate  |                          |
| 4                                                                        | $5.1511 \times 10^{-1}$ | –     | $6.5848 \times 10^{-1}$ | –     | $6.6223 \times 10^{-1}$      | –     | $1.2220 \times 10^{-13}$ |
| 8                                                                        | $2.6813 \times 10^{-1}$ | 0.942 | $3.7678 \times 10^{-1}$ | 0.805 | $3.7965 \times 10^{-1}$      | 0.803 | $5.1946 \times 10^{-13}$ |
| 16                                                                       | $1.3523 \times 10^{-1}$ | 0.988 | $1.8757 \times 10^{-1}$ | 1.006 | $1.8935 \times 10^{-1}$      | 1.004 | $2.0267 \times 10^{-12}$ |
| 32                                                                       | $6.7750 \times 10^{-2}$ | 0.997 | $9.2620 \times 10^{-2}$ | 1.018 | $9.3572 \times 10^{-2}$      | 1.017 | $8.5067 \times 10^{-12}$ |

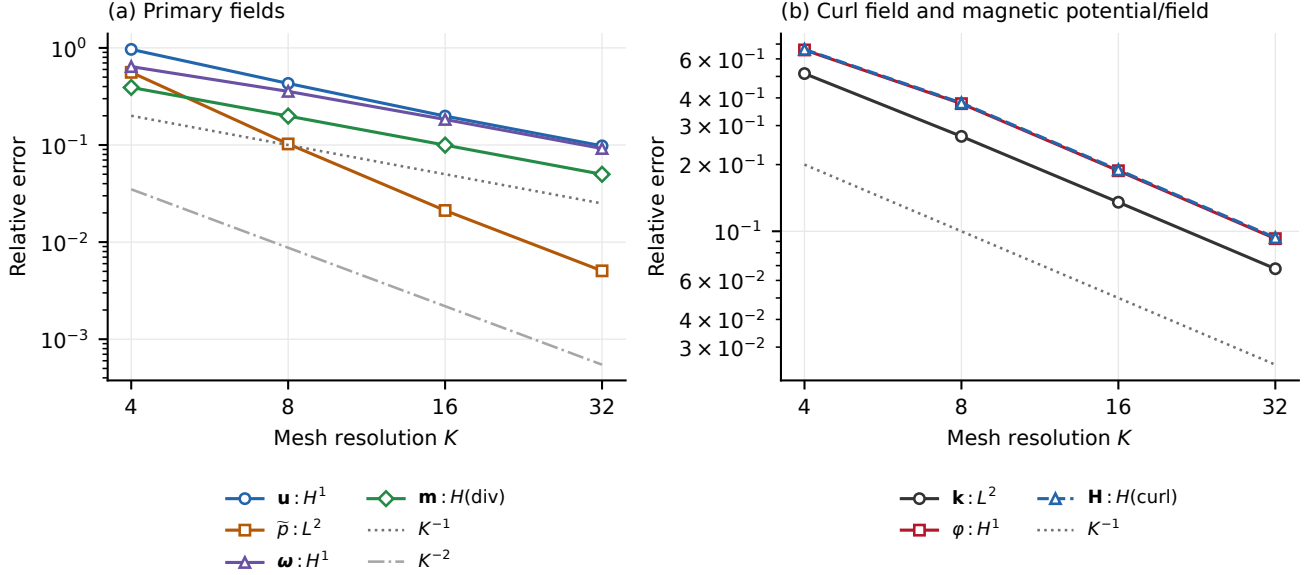


Figure 1: Relative errors for the first-order discretization under joint refinement with  $\Delta t = 1/K$ . (a) Velocity, modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is the mesh resolution  $K = 4, 8, 16, 32$ , so refinement proceeds from left to right. Reference lines proportional to  $K^{-1}$  and  $K^{-2}$  indicate first- and second-order decay with mesh refinement, not independently measured temporal orders.

**Unforced energy dissipation.** We next test free relaxation on the fixed  $K = 8$  mesh, with  $T = 0.5$  and  $\Delta t = 1/16, 1/32, 1/64$ . All physical parameters remain one. Unlike the manufactured-solution test, the applied field and all compensating sources are set to zero:  $\mathbf{H}_e = \mathbf{f} = \mathbf{g} = \mathbf{r} = 0$ . We retain the homogeneous boundary traces described above. The initial shapes are

$$\mathbf{u}_0 = C_u^{-1} \nabla B \times \mathbf{a}, \quad \boldsymbol{\omega}_0 = C_\omega^{-1} B \mathbf{b}, \quad \mathbf{m}_0 = C_m^{-1} S \mathbf{c}.$$

The initial velocity is projected onto the discrete solenoidal space; angular velocity and magnetization are interpolated into their respective spaces. The initial potential is recomputed with zero boundary trace from

$$(\nabla \varphi_h^0, \nabla \psi_h) = -(\mathbf{m}_h^0, \nabla \psi_h) \quad \text{for all admissible } \psi_h,$$

rather than prescribed from the manufactured magnetic potential. The auxiliary variables satisfy their discrete projection relations. Thus the internal magnetic field is not artificially suppressed, and the subsequent solution is not constrained to follow the manufactured temporal profile.

At every time level we integrate the discrete energy

$$E_h^n = \frac{1}{2} (\rho \|\mathbf{u}_h^n\|^2 + \rho \kappa \|\boldsymbol{\omega}_h^n\|^2 + \|\mathbf{m}_h^n\|^2 + \mu_0 \|\nabla \varphi_h^n\|^2). \quad (1)$$

The norms in (1) are spatial  $L^2$  norms. Energy and dissipation are evaluated in the original finite-element spaces using degree-12 quadrature, not from the visualization fields. We also check the balance defect

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^n + N_h^n, \quad N_h^n = \frac{1}{2} \|\mathcal{X}_h^n - \mathcal{X}_h^{n-1}\|_D^2, \quad (2)$$

where  $D_h^n$  is the discrete viscous, spin, magnetic-diffusion, and relaxation dissipation corresponding to  $\mathbf{A}$ . In particular, the weak magnetic diffusion terms are evaluated as  $\sigma \|\mathbf{k}_h^n\|^2$  and  $\sigma \mu_0 \|\vartheta_h^n\|^2$ . For both schemes, the reported balance diagnostic is

$$\epsilon_{\text{bal}}^n = \frac{|R_h^n|}{\max\{E_h^n, E_h^{n-1}, \Delta t D_h^n + N_h^n, 10^{-14} E_h^0\}},$$

with the midpoint dissipation and  $N_h^n = 0$  used for the second-order scheme.

All energy runs use 48 MPI processes with one thread each and the same five-block solver settings as the convergence experiments. The first-order runs require no nonlinear startup. Figure 2 shows monotone decay for all three time steps. The initial discrete energy is approximately  $4.205865 \times 10^{-2}$ ; the final energy ratios and balance diagnostics are given in Table 2. The largest balance diagnostic is  $4.61 \times 10^{-14}$ . These results verify dissipation for the tested discretizations; they do not constitute a pure temporal convergence study.

Table 2: First-order unforced energy test:  $K = 8$ ,  $T = 0.5$ .

| $\Delta t$ | $E_h(T)/E_h(0)$         | $\max_n \epsilon_{\text{bal}}^n$ |
|------------|-------------------------|----------------------------------|
| 1/16       | $9.1847 \times 10^{-5}$ | $3.69 \times 10^{-14}$           |
| 1/32       | $3.1955 \times 10^{-5}$ | $2.22 \times 10^{-14}$           |
| 1/64       | $1.6247 \times 10^{-5}$ | $4.61 \times 10^{-14}$           |

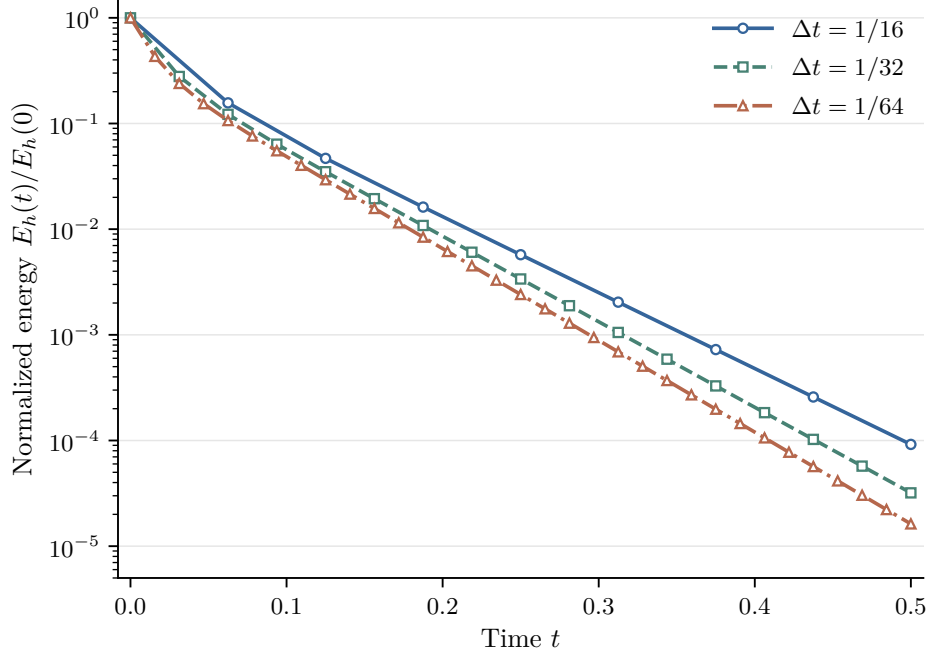


Figure 2: Normalized discrete energy for the first-order scheme in the unforced relaxation test,  $K = 8$  and  $T = 0.5$ . The vertical axis is logarithmic. Symbols represent all computed time levels; connecting lines are not fitted or smoothed curves. The same discrete initial state is used for all three time steps.

## 5 Second-Order Scheme

### 5.3 Numerical results

We apply the second-order discretization to the same manufactured solution, forcing, parameters, and meshes specified in Section 4.3, retaining  $T = 2$  and  $\Delta t = 1/K$ . Velocity and angular velocity now use continuous piecewise quadratic elements, pressure remains continuous piecewise linear, magnetization uses RT1, and  $\mathbf{z}, \mathbf{k}, \mathbf{H}$  use first-family Nedgelec elements one degree above the lowest order. The scalar fields  $\varphi, \vartheta, \varpi$  are continuous piecewise quadratic. The corresponding RT and Nedgelec Basix degree is two.

The first time step is solved by a Broyden quasi-Newton iteration with relative tolerance  $10^{-9}$  and a maximum of 12 iterations. The accepted state is checked by reassembling the nonlinear midpoint equations. Subsequent time steps use the prescribed extrapolated coefficients and require one five-block linear solve per step. The  $\mathbf{z}_h$  and  $\varpi_h$  slots represent the current midpoint-stage projections; they are not averaged again with the preceding stage.

The primary fields are compared at  $T$ . The pressure slot represents the averaged modified pressure, so its reference is

$$\tilde{p}_{\text{ref}}^{N-1/2} = \frac{\tilde{p}(T) + \tilde{p}(T - \Delta t)}{2},$$

with the same zero-mean convention as above. This reference is not identified with either the endpoint pressure or the analytic pressure evaluated directly at the midpoint.

Table 3 and Fig. 3 show that the finest-grid rates for  $\mathbf{u}, \tilde{p}, \boldsymbol{\omega}, \mathbf{m}, \varphi, \mathbf{H}$  are 2.166, 2.051, 1.981, 2.007, 1.968, and 1.968, respectively. The  $L^2$  error of  $\mathbf{k}$  decreases approximately at first order, with a finest-grid rate of 1.015; second-order convergence is therefore not claimed for every auxiliary quantity. The sampled curl diagnostic ranges from  $7.30 \times 10^{-13}$  to  $6.96 \times 10^{-11}$ . For  $K = 32$ , all 64 steps reach  $T = 2$ , and the final full-system relative algebraic residual is  $2.08 \times 10^{-13}$ . These observations support the reported joint-refinement behavior. Because the mesh and time step vary together and the primary exact fields have linear temporal amplitudes, they do not constitute a separate test of pure temporal accuracy or of energy stability.

Table 3: Relative errors and observed rates for the second-order scheme,  $T = 2$  and  $\Delta t = 1/K$ .  $C_H$  is the sampled absolute curl diagnostic.

| (a) Velocity, pressure, angular velocity, and magnetization              |                         |       |                         |       |                              |       |                             |       |
|--------------------------------------------------------------------------|-------------------------|-------|-------------------------|-------|------------------------------|-------|-----------------------------|-------|
| $K$                                                                      | $\mathbf{u}: H^1$       |       | $\tilde{p}: L^2$        |       | $\boldsymbol{\omega}: H^1$   |       | $\mathbf{m}: H(\text{div})$ |       |
|                                                                          | Error                   | Rate  | Error                   | Rate  | Error                        | Rate  | Error                       | Rate  |
| 4                                                                        | $5.6873 \times 10^{-1}$ | —     | $4.2000 \times 10^{-1}$ | —     | $2.1381 \times 10^{-1}$      | —     | $1.1937 \times 10^{-1}$     | —     |
| 8                                                                        | $1.5090 \times 10^{-1}$ | 1.914 | $9.4815 \times 10^{-2}$ | 2.147 | $6.2017 \times 10^{-2}$      | 1.786 | $3.1613 \times 10^{-2}$     | 1.917 |
| 16                                                                       | $2.4805 \times 10^{-2}$ | 2.605 | $2.0872 \times 10^{-2}$ | 2.184 | $1.6320 \times 10^{-2}$      | 1.926 | $7.8383 \times 10^{-3}$     | 2.012 |
| 32                                                                       | $5.5259 \times 10^{-3}$ | 2.166 | $5.0382 \times 10^{-3}$ | 2.051 | $4.1345 \times 10^{-3}$      | 1.981 | $1.9494 \times 10^{-3}$     | 2.007 |
| (b) Auxiliary curl field, potential, magnetic field, and curl diagnostic |                         |       |                         |       |                              |       |                             |       |
| $K$                                                                      | $\mathbf{k}: L^2$       |       | $\varphi: H^1$          |       | $\mathbf{H}: H(\text{curl})$ |       | $C_H$                       |       |
|                                                                          | Error                   | Rate  | Error                   | Rate  | Error                        | Rate  | Error                       | Rate  |
| 4                                                                        | $2.1008 \times 10^{-1}$ | —     | $2.0119 \times 10^{-1}$ | —     | $2.0303 \times 10^{-1}$      | —     | $7.3001 \times 10^{-13}$    | —     |
| 8                                                                        | $9.5804 \times 10^{-2}$ | 1.133 | $6.0098 \times 10^{-2}$ | 1.743 | $6.0718 \times 10^{-2}$      | 1.742 | $3.9064 \times 10^{-12}$    | —     |
| 16                                                                       | $4.6084 \times 10^{-2}$ | 1.056 | $1.6119 \times 10^{-2}$ | 1.899 | $1.6289 \times 10^{-2}$      | 1.898 | $1.8492 \times 10^{-11}$    | —     |
| 32                                                                       | $2.2804 \times 10^{-2}$ | 1.015 | $4.1189 \times 10^{-3}$ | 1.968 | $4.1627 \times 10^{-3}$      | 1.968 | $6.9611 \times 10^{-11}$    | —     |

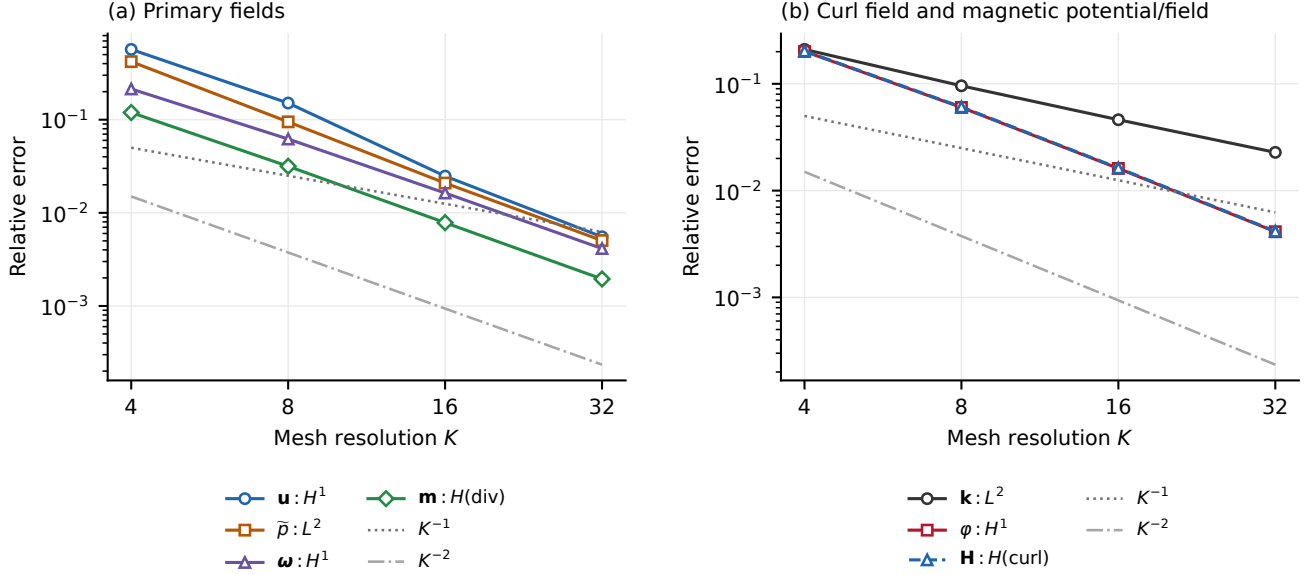


Figure 3: Relative errors for the second-order discretization using the same manufactured solution and  $\Delta t = 1/K$ . (a) Velocity, averaged modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is  $K = 4, 8, 16, 32$ , increasing from left to right. Reference lines proportional to  $K^{-1}$  and  $K^{-2}$  distinguish the nearly quadratic decay of the principal errors from the approximately first-order  $L^2$  decay for  $\mathbf{k}$ .

**Unforced energy dissipation.** We repeat the zero-field, zero-source relaxation test with  $K = 8$ ,  $T = 0.5$ , and  $\Delta t = 1/16, 1/32, 1/64$ , using the same continuous initial shapes and constraint-compatible initialization as in the first-order energy test. The second-order finite-element spaces specified above are retained. Consequently, the two schemes do not share an identical discrete initial state; each scheme does use the same initial state for all three time steps. For this energy test, the initial Broyden solve uses relative tolerance  $10^{-10}$  with at most 20 iterations, and the accepted midpoint equations are independently reassembled to check their residual. Subsequent time steps use the prescribed extrapolation.

The energy is evaluated by (1). For the second-order discretization, the balance defect is

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^{n-1/2}, \quad (3)$$

where  $D_h^{n-1/2}$  is evaluated at the midpoint averages of the primary fields and the endpoint auxiliaries  $\mathbf{k}_h, \vartheta_h$ . There is no first-order numerical-dissipation term  $N_h^n$ . The stage fields  $\mathbf{z}_h, \varpi_h$  are not averaged a second time.

Figure 4 shows monotone energy decay for all three time steps, with initial energy approximately  $4.295033 \times 10^{-2}$ . Table 4 reports the final ratios and balance diagnostics; the largest normalized balance defect is  $3.71 \times 10^{-11}$ . The coarsest time step produces slower late-time decay, which is retained in the plot. Energy dissipation therefore should not be interpreted as evidence that the temporal error is negligible. Since the two schemes also use different spatial approximation orders, these figures are separate stability tests rather than a controlled comparison of temporal accuracy. The computations provide numerical evidence of the discrete dissipation mechanism at the tested step sizes, not a proof for arbitrary step sizes.

Table 4: Second-order unforced energy test:  $K = 8$ ,  $T = 0.5$ .

| $\Delta t$ | $E_h(T)/E_h(0)$         | $\max_n \epsilon_{\text{bal}}^n$ |
|------------|-------------------------|----------------------------------|
| 1/16       | $1.1117 \times 10^{-4}$ | $3.71 \times 10^{-11}$           |
| 1/32       | $2.5984 \times 10^{-5}$ | $7.77 \times 10^{-13}$           |
| 1/64       | $1.0438 \times 10^{-5}$ | $2.67 \times 10^{-13}$           |

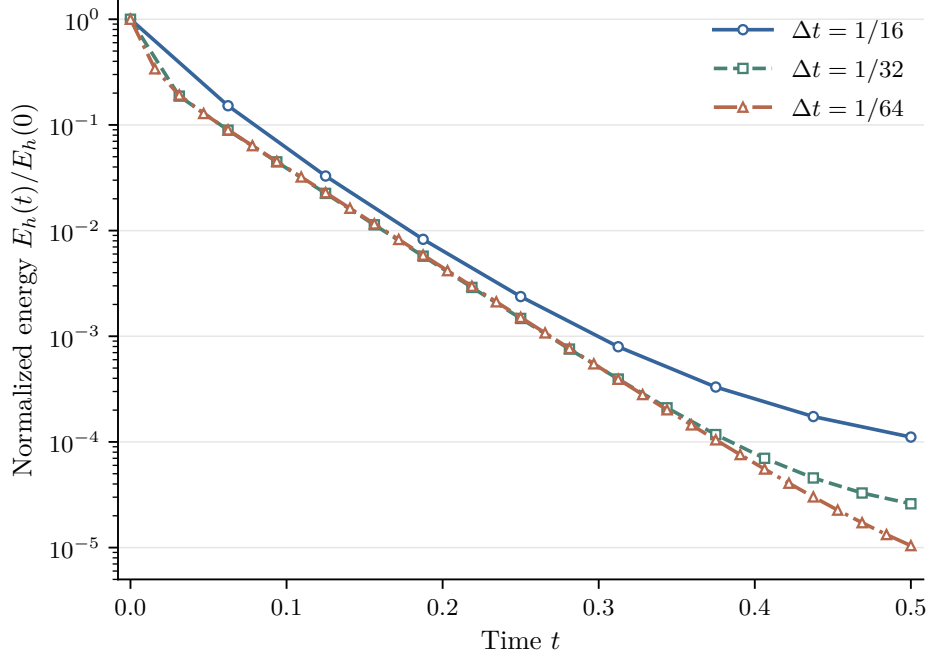


Figure 4: Normalized discrete energy for the second-order scheme in the unforced relaxation test,  $K = 8$  and  $T = 0.5$ . The logarithmic vertical scale matches Figure 2. Symbols represent all computed time levels without smoothing. The same discrete initial state is used for all three time steps.

## References

- [1] I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime, and G. N. Wells, DOLFINx: The next generation FEniCS problem solving environment, 2023.